Gradient Descent

Trust the line, one small step at a time

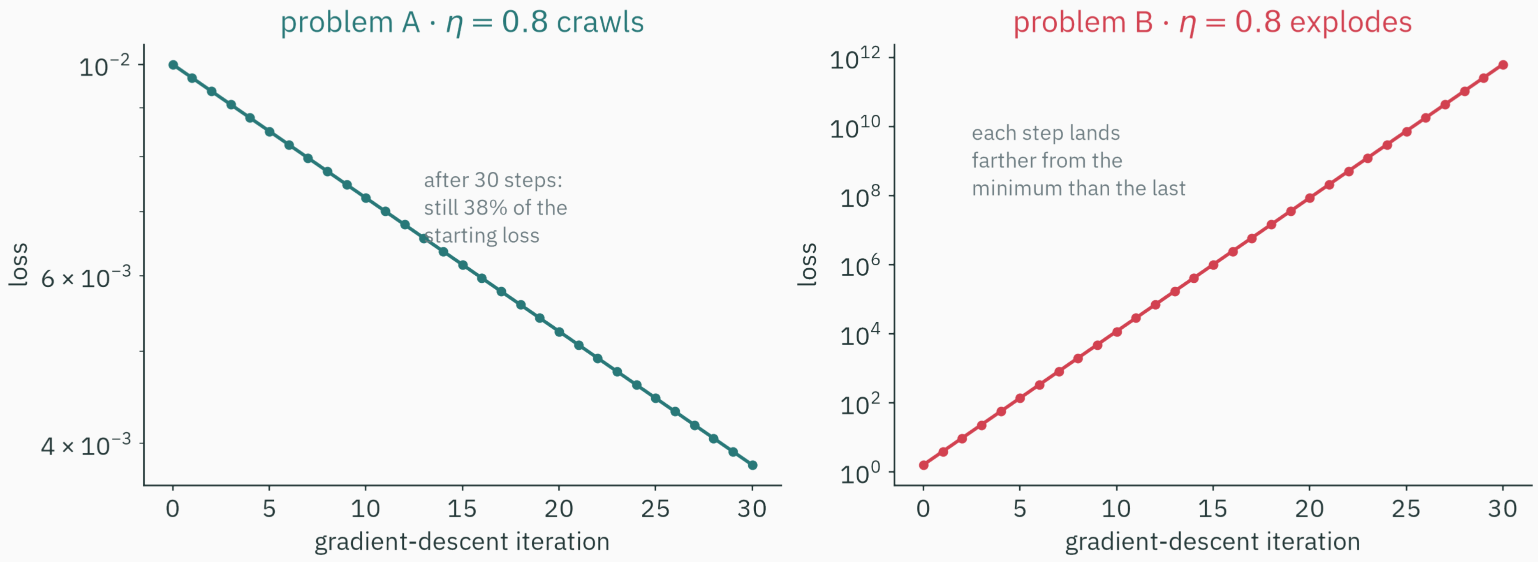
Prof. Nipun Batra

Mathematical Foundations for AI · IIT Gandhinagar

The hook: Mystery 2 comes home

Same code, same η — two fates

L1, Mystery 2. Gradient descent, same code, same learning rate $\eta = 0.8$ — on two different problems:



It crawls on one and explodes on the other. Sixteen lectures later, we can finally arrest someone. **Today this mystery is resolved completely** — with exact numbers.

Interrogating the suspect

The obvious suspect is $\eta = 0.8$. Put it on the stand:

Hypothesis	**Then we'd expect...**	**Verdict**
η is too **big**	both problems overshoot and explode	× A crawls
η is too **small**	both problems crawl	× B explodes

The same number cannot be simultaneously too big and too small...

...unless "big" and "small" are measured relative to something in the problem. Naming that something is today's whole job.

Everything we need is already on the table

from	what it gave us
L7	the local line: $f(x_0+h) \approx f(x_0) + f'(x_0)h$ — “L17 will quote today’s line”
L8	the direction: $-\nabla f$ is steepest descent (Cauchy–Schwarz) — we even ran a small GD
L16	the license: on convex bowls, local information is globally trustworthy

- L8 settled **which way** to step. Nobody has yet said **how far**.
- Today: derive the algorithm, then interrogate its one knob — η — until it confesses.

Learning outcomes

By the end of this lecture you will be able to:

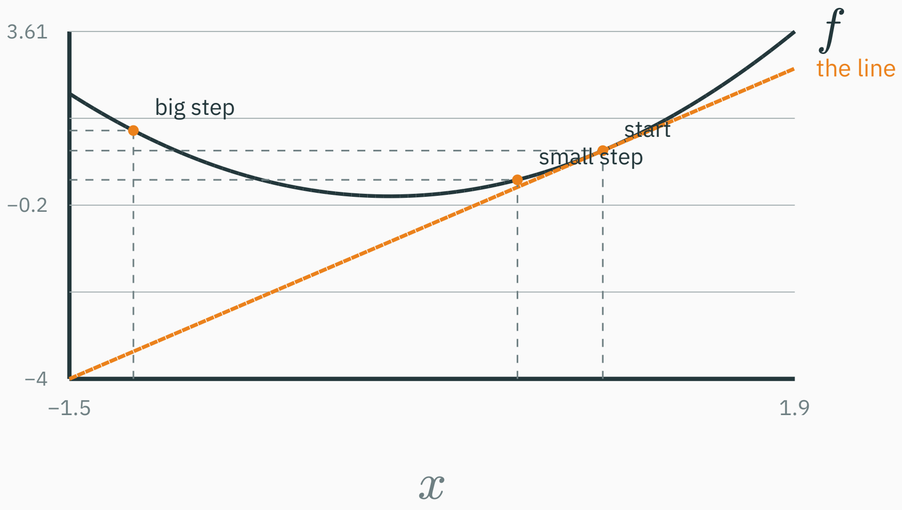
1. **Derive** gradient descent from the first-order Taylor approximation (the L7 payoff).
2. Implement the **four-line algorithm** and read η as a trust radius.
3. Predict **crawl / converge / oscillate / diverge** from η and the curvature — the wall at $\eta = 2/a$.
4. Run the **exact analysis on quadratics**: contraction factor $1 - \eta a$, per-axis rates.
5. Compute a **condition number** κ and explain the zig-zag pathology it causes.
6. Explain why **feature scaling** speeds up training, and how **momentum** fights ravines.

Trust the line: deriving the algorithm

The one idea

V VISUAL

Stand at $x = 1$ on $f(x) = x^2$. The derivative hands you a line — and the line makes a **promise**: "walk downhill along me and f keeps dropping."



Small step: the promise is kept. Big step: the line says "still descending!" while the truth ($f = 1.44$) is **above where you started**. Lines are honest only **locally**.

How far can you trust a line?

L7's local replacement, in today's many-variable clothes (L9): for a step Δ from θ,

$$f(\theta + \Delta) \approx f(\theta) + \nabla f(\theta)^\top \Delta$$

- The error of this replacement grows like $\|\Delta\|^2$ (the second-order Taylor term) — **double** your step, the line lies **four times** harder.
- So every use of the line comes with a **radius of trust**: a $\|\Delta\| \leq r$ inside which the promise roughly holds.

This is the spine of the lecture: **trust the linear model for one small step, then rebuild the line and repeat**. Everything else is bookkeeping about r.

Which direction? L8 already answered

Inside the trust radius, pick the step the **line** likes best:

$$\min_{\|\Delta\| \le r} \ \nabla f(\theta)^\top \Delta$$

L8 callback — by Cauchy–Schwarz the dot product is most negative when Δ points **opposite** the gradient:

$$\Delta^\star = -r \ \frac{\nabla f(\theta)}{\|\nabla f(\theta)\|}$$

- Direction: $-\nabla f$, dead against the gradient — **steepest descent**, exactly L8′s compass.
- New today: the length r — the part L8 left open.

The update rule

Fold the radius and the gradient's length into one knob, $\eta > 0$:

$$\boxed{\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t)}$$

- η is the **learning rate** — literally the size of the trusted step.
- Rebuild the line at the new point, step again. That is the entire method.

Cash the L7 promissory note: this update "quotes today's local line" — L7′s closing table predicted this exact formula would appear here.

Every trusted step pays out

Substitute the step $\Delta = -\eta\nabla f$ back into the linear model:

$$f(\theta - \eta\nabla f) \approx f(\theta) - \eta\|\nabla f(\theta)\|^2$$

- $\|\nabla f\|^2 \geq 0$: while the line holds, **every step decreases f** — by more where the surface is steeper.
- Near a minimum $\nabla f \to 0$: the payout shrinks, steps shrink, the walk settles **by itself**.

The fine print: the payout is promised **by the line**. Step past the trust radius and the true f can go **up** — we watched it happen on the previous figure.

The whole algorithm is four lines

```
theta = theta0                    # 1. start somewhere
for t in range(T):                # 2. repeat
    g = grad(L, theta)            # 3. which way is uphill  (L8)
    theta = theta - lr * g        # 4. small trusted step downhill
```

- Stop when $\|g\|$ is tiny, the loss plateaus, or the budget T runs out.
- This loop — with refinements — trains **everything**: linear regression, GPT, the lot.

One hyperparameter, no memory, only first derivatives. The rest of today: what that one hyperparameter does.

Numbers first: halving your way down

$f(x) = x^2$ (curvature $f'' = a = 2$), learning rate $\eta = 0.25$, start $x_0 = 2$:

$$x_1 = 2 - 0.25 \cdot (2 \cdot 2) = 1, \quad x_2 = 1 - 0.25 \cdot 2 = 0.5, \quad x_3 = 0.25, \quad \ldots$$

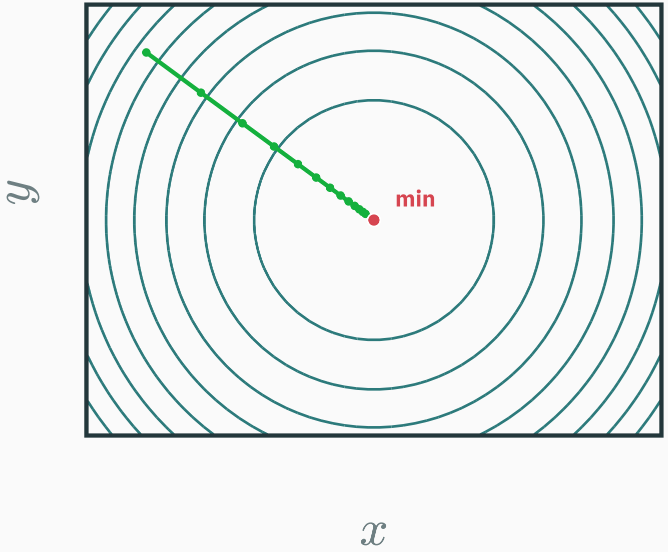
- Every step lands exactly **halfway** to the minimum: $x_{t+1} = x_t - 0.25 \cdot 2x_t = 0.5\, x_t$.
- The distance to the minimum shrinks by the **same factor 0.5 every step** — a clean geometric decay.

Hold that thought. That per-step factor is about to become the main character of the lecture.

On a round bowl, GD is unstoppable

V VISUAL

$f(x, y) = x^2 + y^2$ — same curvature in every direction:



round contours $\Rightarrow -\nabla f$ points **straight at the minimum**; steps shrink as $\|\nabla f\| \to 0$

Remember this picture — today's villain will bend it out of shape.

What η really is

- **Too timid**: you trust the line less than you could — you collect only a sliver of the guaranteed payout per step.
- **Too bold**: you leave the trust region — the line's promise is void, and the true f may rise.
- **Just right**: the largest step the line can still honor.

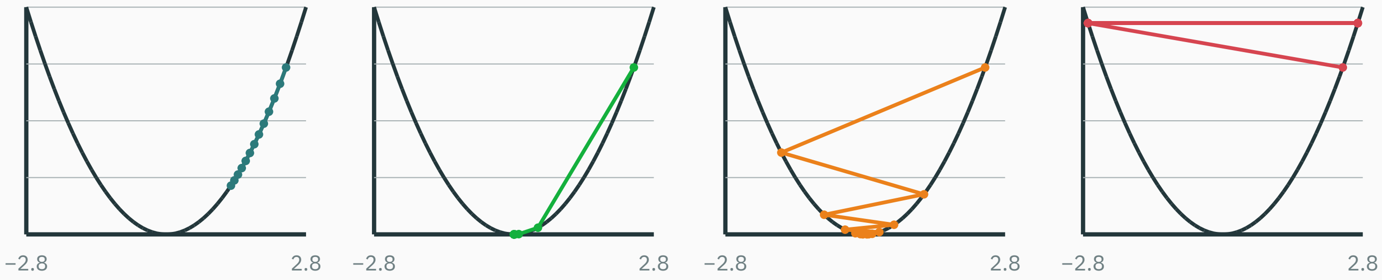
How large is that, exactly? It must depend on how fast the line goes stale — the **curvature**. That is a number we can compute. Next section: compute it.

The learning-rate dial

One knob, four personalities

GD on $f(x) = \frac{1}{2}x^2$ (curvature $a = 1$), $x_0 = 2.4$, twelve steps — four choices of η:

$\eta = 0.05$ · **crawls** $\eta = 0.8$ · **converges** $\eta = 1.7$ · **oscillates in** $\eta = 2.15$ · **diverges**

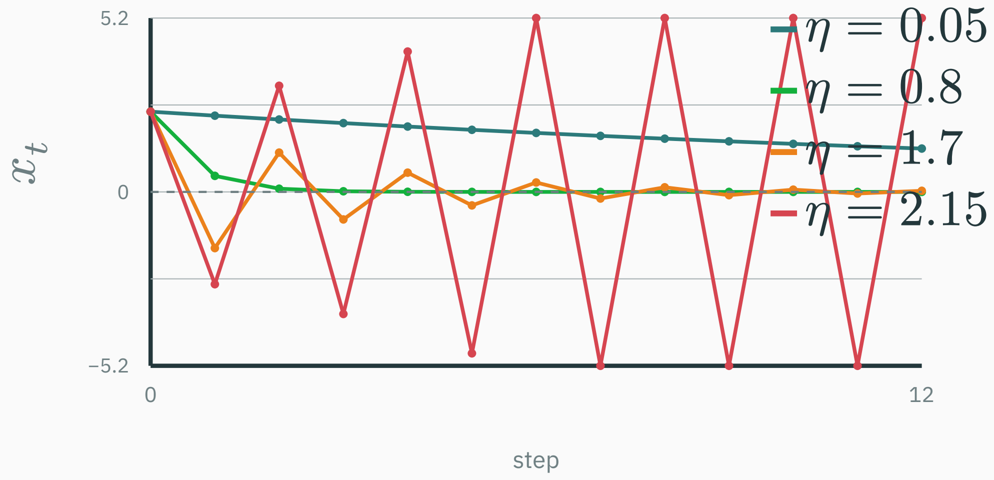


every dot is a real iterate, computed by this slide — crawl · converge · overshoot-but-converge · explode

The same four walks, in time

V VISUAL

Plot the iterate x_t itself against the step count:



- crawl and converge stay on one side; overshoot **crosses the minimum every step** and still converges; diverge crosses and **grows** (clipped at ± 5.2).

The wall sits at exactly $\eta = 2/a$

On a bowl of curvature a, the step from x has length $\eta a\,|x|$ — against a distance-to-minimum of $|x|$:

- $\eta a < 1$: land **short** of the minimum — same side, monotone approach.
- $1 < \eta a < 2$: land **past** the minimum but **closer** than you started — the overshooting dance.
- $\eta a > 2$: land past it and **farther out** — each step amplifies the error. Divergence.

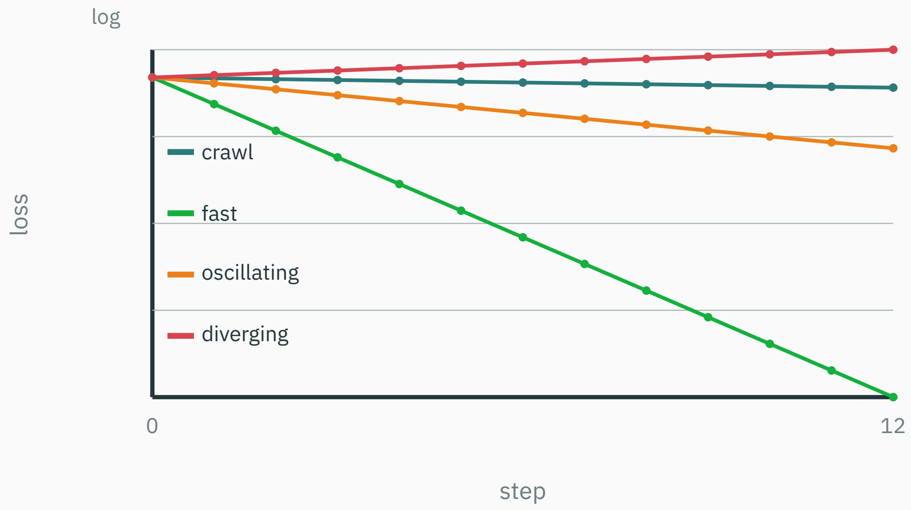
The speed limit is $\eta < 2/a$ — two divided by the curvature.

For a general smooth loss, the worst curvature anywhere is called L ("L-smooth" in every optimization text) — the classical safe range is $\eta < 2/L$. Same wall, bigger vocabulary.

What you'll actually see: the training curve

V VISUAL

In 10^6 dimensions you can't watch x_t — only the **loss**. Same four runs, log scale (L2 taught us to love logs):



geometric decay = **straight line** on a log axis; the slope is the per-step factor — divergence slopes **up**

You will diagnose learning rates from curves like these for the rest of your career.

Interactive: sweep the dial yourself

INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/optimizer-race

Drop a start point on a 2-D loss surface and sweep the learning rate live: watch the same four personalities — crawl, converge, overshoot, explode — appear as you cross the $2/a$ wall.

Find the wall experimentally on the bowl, then check it against the curvature readout.

Checkpoint: the knife edge

Q QUESTION

Run GD on $f(x) = (x-3)^2$ — curvature $a = f'' = 2$ — with $\eta = 1$, from $x_0 = 0$. What happens?

A. Converges to $x = 3$ in one step

B. Diverges to infinity

C. Bounces between 0 and 6 forever

D. Converges slowly, like 0.5^t

Answer: the knife edge

A ANSWER

Correct: C — bounces between 0 and 6 forever

Why: The update is $x \leftarrow x - 2(x-3) = 6 - x$: each step **reflects** through the minimum. $\eta = 1$ is exactly the wall $2/a = 1$ — the error factor is -1: never shrinks, never grows.

$$x: \quad 0 \to 6 \to 0 \to 6 \to \cdots \quad \text{— a perpetual, loss-preserving dance around } x^* = 3$$

The exact story on quadratics

Why quadratics tell the whole story

Zoom in near any minimum θ^* of any smooth loss (L7′s hook, L9′s Taylor):

$$f(\theta) \approx f(\theta^*) + \frac{1}{2}\, f''(\theta^*)(\theta - \theta^*)^2$$

- The endgame of **every** training run is played on an approximate quadratic — curvature $a = f''(\theta^*)$.
- And on a quadratic we don't have to approximate anything: GD can be solved **exactly**, in one line.

Exact answers on quadratics = the local theory of gradient descent on everything.

⭐ One line of algebra

D DERIVATION

Take $f(x) = \frac{a}{2}\,x^2$ with curvature $a > 0$, so $f'(x) = ax$. Apply the update:

$$x_{t+1} = x_t - \eta\ ax_t = \underbrace{(1 - \eta a)}_{\text{one number}}\ x_t$$

Unroll it — the whole future in closed form:

$$x_t = (1 - \eta a)^t\ x_0$$

GD on a quadratic is repeated multiplication by the constant $1 - \eta a$.
Nothing else happens.

Check against the halving demo: $a = 2$, $\eta = 0.25 \implies 1 - \eta a = 0.5$. ✓

⭐ The contraction factor decides everything

D DERIVATION

Give $c = 1 - \eta a$ its name: the **contraction factor**. Error after t steps: $|c|^t\ |x_0|$; loss shrinks by c^2 per step.

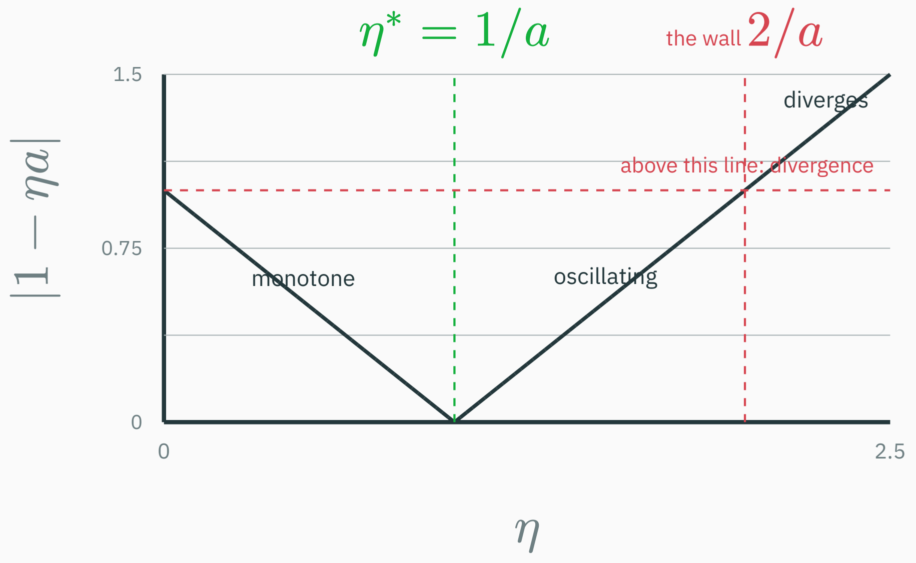
η range	**factor c**	**behavior**
$0 < \eta < 1/a$	$0 < c < 1$	monotone convergence (crawl if $c \approx 1$)
$\eta = 1/a$	$c = 0$	the minimum in **one step**
$1/a < \eta < 2/a$	$-1 < c < 0$	overshoots each step, still converges
$\eta = 2/a$	$c = -1$	bounces forever (the knife edge)
$\eta > 2/a$	$c < -1$	diverges

Convergence $\Leftrightarrow |1 - \eta a| < 1 \Leftrightarrow 0 < \eta < 2/a$. The best η is $1/a$ — one over the curvature.

The factor, as a picture

V VISUAL

$|1-\eta a|$ against η (drawn for $a=1$):



a V with its tip at $\eta = 1/a$: undershoot on the left branch, overshoot on the right, cross $|c| = 1$ and you burn

Mystery 2 · the numbers confess

D DERIVATION

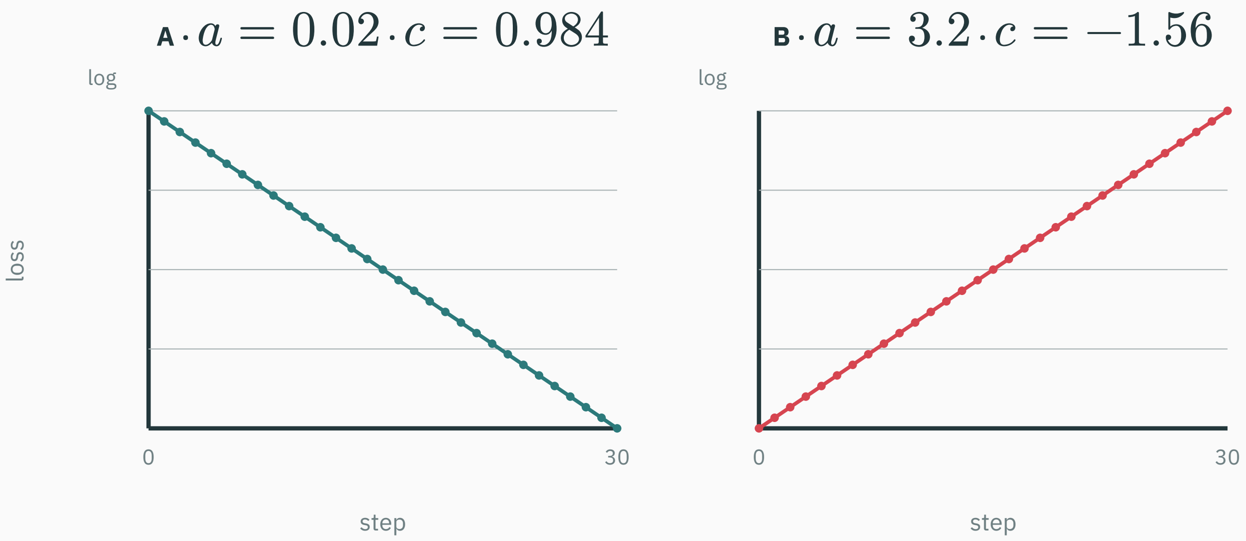
Confession time. L1′s two “problems” were exactly $f(x) = \frac{a}{2}x^2$ — with different curvatures, same $\eta = 0.8$:

	problem A · $a = 0.02$	**problem B** · $a = 3.2$
factor $c = 1 - 0.8a$	0.984	-1.56
$\lvert c \rvert$ vs 1	just below — barely contracts	above — **expands**
after 30 steps	$0.984^{30} = 0.62$ of the error left $\implies$ loss $\times\, 0.38$	error $\times\, 1.56^{30}$; loss $\times\, 2.43$ per step $\approx 10^{11}$

- L1′s figure annotated problem A with “still 38% of the starting loss” — that number is 0.984^{60}. **Predicted by one line of algebra.**
- For problem B, $\eta = 0.8$ sits beyond its wall $2/3.2 = 0.625$. It never had a chance.

Mystery 2 · recomputed before your eyes

The L1 figure again — except now **this slide computes it** from $x_{t+1} = (1 - \eta a)x_t$:



straight lines on the log axis — geometric decay and geometric explosion, slopes $2\log|c|$

Mystery 2, case closed: "too small" and "too big" were both true — of the same η, against two different curvatures.

⭐ Two axes, one dial

D DERIVATION

The deeper trouble: one problem can contain **both curvatures at once**. Take the diagonal bowl

$$f(x, y) = \frac{1}{2}(ax^2 + by^2), \quad \nabla f = (ax, by), \quad a \text{ flat}, \ b \text{ steep}$$

The update never mixes the coordinates — GD **decouples** into two 1-D stories with one shared η:

$$x_t = (1 - \eta a)^t x_0, \qquad y_t = (1 - \eta b)^t y_0$$

- Stability is hostage to the **steep** axis: need $\eta < 2/b$.
- Speed is hostage to the **flat** axis: its factor $1 - \eta a$ then sits near 1.

⭐ Two axes, one dial

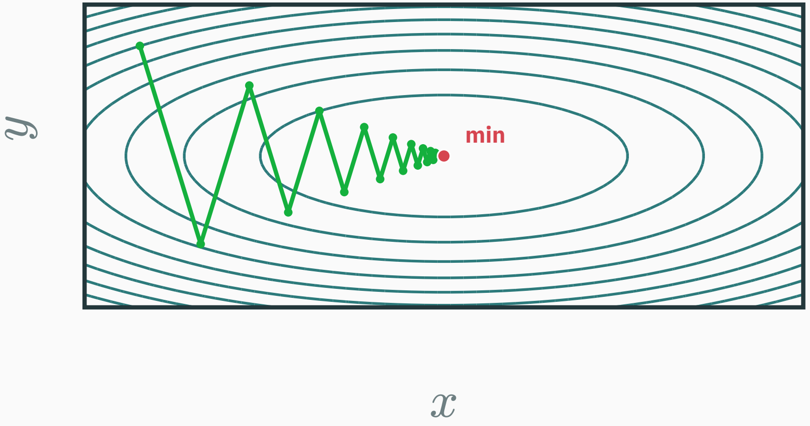
D DERIVATION

General bowls? L5 callback: eigenvectors of the Hessian rotate any bowl into this diagonal form — a and b are just its extreme eigenvalues. The diagonal case **is** the general case.

Watch both factors at once

V VISUAL

$f(x, y) = \frac{1}{2}(x^2 + 9y^2)$, $\eta = 0.2$: per-axis factors $1 - 0.2 \cdot 1 = 0.8$ and $1 - 0.2 \cdot 9 = -0.8$.



y flips sign every step (factor -0.8) while x shrinks steadily (factor $+0.8$) — both visible in one real trajectory

No η serves two curvatures

Try to pick one η for A's and B's curvatures **simultaneously** ($a = 0.02$, $b = 3.2$):

η	**flat axis ($a = 0.02$)**	**steep axis ($b = 3.2$)**
0.8 (L1′s)	$c = 0.984$ — crawls	$c = -1.56$ — explodes
$0.625 = 2/b$	$c = 0.9875$ — crawls	$c = -1$ — bounces forever
0.5	$c = 0.99$ — crawls **worse**	$c = -0.6$ — fine
$0.3125 = 1/b$	$c = 0.99375$ — crawls worse still	$c = 0$ — one step!
$50 = 1/a$	$c = 0$ — one step!	$c = -159$ — catastrophic

Every row sacrifices an axis. When curvatures are far apart, no learning rate is good — that situation has a name.

The villain's name: the condition number

D DERIVATION

$$\kappa = \frac{a_{\max}}{a_{\min}} \quad \text{(steepest curvature over flattest — for Mystery 2's pair: } 3.2/0.02 = 160\text{)}$$

Best compromise: $\eta^* = 2/(a_{\max} + a_{\min})$ balances the two factors at

$$|c| = \frac{\kappa - 1}{\kappa + 1} \approx 1 - \frac{2}{\kappa}$$

κ	1	10	160	10^4
best factor	0	0.818	0.9876	0.9998
steps to cut error $100\times$	1	≈ 23	≈ 369	$\approx 23,000$

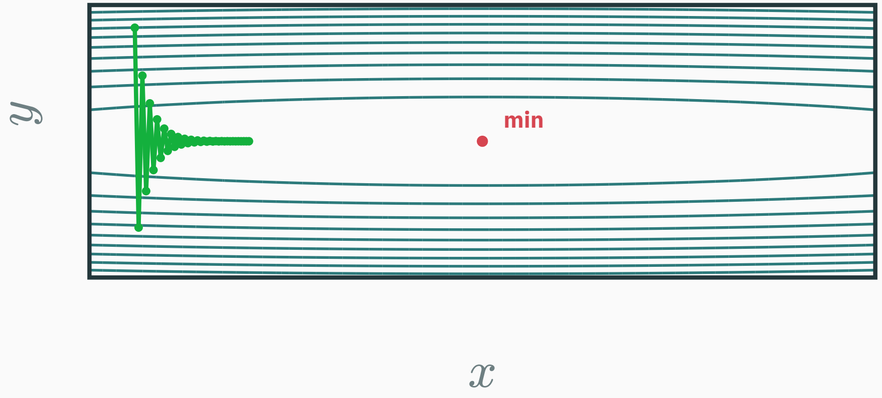
GD's bill grows in proportion to κ. Conditioning — not dimension — decides your fate.

The ill-conditioned bowl

Both personalities in one problem

V VISUAL

Put Mystery 2′s two curvatures into **one** bowl: $f(x,y) = \frac{1}{2}(0.02x^2 + 3.2y^2)$, $\kappa = 160$ — and run GD with a safe $\eta = 0.55$:



the infamous **zig-zag**: oscillate across the ravine (factor -0.76), crawl along it (factor 0.989)

A real loss over 10^6 parameters has 10^6 curvatures. Its κ — worst over best — routinely reaches the thousands. Every training run lives inside this picture.

Why zig-zag? L8 told us

- L8 ⭐: every GD step is **perpendicular to the contour** it starts on.
- In a ravine, contours are stretched ellipses — perpendicular-to-the-contour points **across** the valley, hardly **along** it.
- The steep axis pins η below its wall $2/b$, so the along-valley axis moves at factor $1 - \eta a \approx 1 - 2/\kappa$: a crawl.

Zig-zag is not a bug in the code — it is steepest descent doing exactly what "steepest" means, on a bowl where steepest ≠ toward the minimum.

Where do squashed bowls come from? Your features

Fit a price from two features by least squares. The loss curvature along weight w_j is $a_j = \text{mean}\left(x_j^2\right)$ (the Hessian is $\frac{1}{n}X^\top X$ — L16′s matrix):

feature	**values**	**curvature** $\text{mean}\left(x_j^2\right)$
house area (m²)	$80, 120, 160$	$46400/3 \approx 15,467$
rooms	$2, 3, 4$	$29/3 \approx 9.67$

$$\kappa = \frac{46400}{29} = 1600$$

Nobody chose an ill-conditioned bowl. It walked in with the units of the data — curvature is the feature scale, squared.

The fix costs one line: standardize

```
X = (X - X.mean(axis=0)) / X.std(axis=0)   # each column: mean 0, std 1
```

- Every column now has $\mathrm{mean}(x_j^2) = 1 \implies$ all curvatures comparable $\implies \kappa \approx 1$: the bowl is **round again**, and one η serves all axes.
- This is why `StandardScaler` opens almost every ML pipeline: it is not cosmetics — it **reshapes the loss surface**. The fancy word is **preconditioning**.

Deep nets can't pre-scale their internal activations by hand — so they bolt the same idea inside the network: BatchNorm / LayerNorm. Story continued in ES 667.

Checkpoint: units and curvature

Q QUESTION

You re-measure the "area" feature in centimeters² instead of m² — every value grows $\times 10^4$. In the least-squares loss, the curvature along that weight's axis becomes...

A. unchanged — units can't matter

B. $\times 10^4$

C. $\times 10^8$

D. $\div 10^4$

Answer: units and curvature

A ANSWER

Correct: C — $\times\, 10^8$

Why: Curvature along w_j is $\mathrm{mean}(x_j^2)$ — the feature scale enters **squared**: $(10^4)^2 = 10^8$. One careless unit choice, eight orders of magnitude of conditioning damage (and the safe η shrinks by 10^8 to match).

Same lesson from the other side: the **optimum weight** shrinks by 10^4 — the minimum didn't move in "price space", but the bowl got squashed 10^8-fold along that axis.

Momentum: memory against the ravine

A marble doesn't zig-zag

Roll a heavy marble down the ravine instead of teleporting a point:

- The gradient acts as a **force**; the marble accumulates **velocity**.
- Across the valley the force flips direction every instant — the pushes **cancel** in the velocity.
- Along the valley the force always agrees — the pushes **add up**, and the marble accelerates.

Same information (∇f), one new state variable (velocity) — and the ravine's two personalities get opposite treatment. This is the heavy-ball method.

The update rule: two lines now

$$v_{t+1} = \beta\, v_t + \nabla f(\theta_t), \qquad \theta_{t+1} = \theta_t - \eta\, v_{t+1}$$

```
v = 0
for t in range(T):
    g = grad(L, theta)
    v = beta * v + g            # remember where we've been going
    theta = theta - lr * v      # step along the memory, not the gradient
```

- $\beta \in [0, 1)$ is the **momentum coefficient** — how much velocity survives each step. Default: 0.9.

Unroll it: a moving average of gradients

Feed the recursion into itself:

$$v_t = g_t + \beta g_{t-1} + \beta^2 g_{t-2} + \beta^3 g_{t-3} + \cdots$$

- A **geometrically-weighted memory** of every gradient so far: recent ones loud, old ones fading by β each step.
- GD is the $\beta = 0$ special case: memory of exactly one gradient.

You have met this object: it is a geometric series in disguise — the same $1 + \beta + \beta^2 + \cdots = 1/(1-\beta)$ that priced L2′s float gaps and L15′s discounted pseudo-counts.

Numbers: cancel vs accumulate

$\beta = 0.9$. Compare the two axes of the ravine, in steady state:

Along the valley — gradients keep agreeing, say $g = -0.4$ every step:

$$v \to -0.4(1 + \beta + \beta^2 + \cdots) = \frac{-0.4}{1 - 0.9} = -4 \quad \text{— amplified} \times 10$$

Across the valley — gradients alternate $+2, -2, +2, \ldots$:

$$v \to 2(1 - \beta + \beta^2 - \cdots) = \frac{2}{1 + 0.9} \approx 1.05 \quad \text{— damped to half}$$

Momentum is a direction filter: consistent signals gain $1/(1-\beta)$, flip-flopping signals shrink to $1/(1+\beta)$.

How much memory? $1/(1-\beta)$ steps

β	**effective memory**	**character**
0.5	≈ 2 steps	barely more than GD
0.9	≈ 10 steps	the workhorse default
0.99	≈ 100 steps	a supertanker — smooth, slow to turn

- Along consistent directions the **effective** step is $\approx \eta/(1-\beta)$ — budget η accordingly (that's why the momentum run below uses a smaller η).

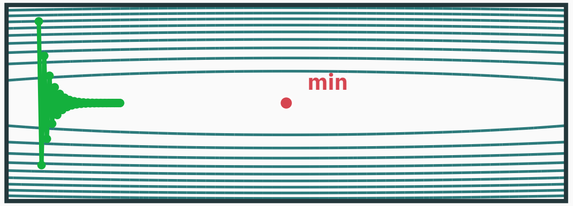
Too much memory overshoots: the marble sails past the valley's end and **rings** back and forth. If the loss oscillates on a slow timescale, suspect β, not η.

The ravine, rerun

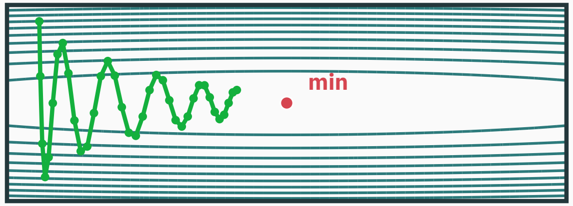
V VISUAL

Same bowl ($\kappa = 160$), same start, same 36 steps:

plain GD · $\eta = 0.55$



momentum · $\eta = 0.21, \beta = 0.9$



the cross-valley wobble is filtered out; the along-valley coordinate — GD's crawl — actually **arrives**

Momentum doesn't shrink κ — it changes how much κ hurts.

Interactive: the momentum map

I INTERACTIVE

INTERACTIVE

↗ distill.pub/2017/momentum/

Distill, **“Why Momentum Really Works”** — this week’s assigned reading, and the most beautiful interactive article in optimization. Drag η and β across the 2-D stability map; watch the ravine’s per-axis rates respond exactly as today’s algebra predicts.

As you read, test yourself: the article’s “effective step size” $\eta/(1-\beta)$, its condition-number plots, its ringing regime — every one of them appeared on a slide today.

Summary & what’s next

The whole lecture in one table

symptom (loss curve)	diagnosis	medicine
barely-tilted straight decay	$\eta a_{\min} \ll 1$: crawling	raise η toward the wall
loss rises geometrically	$\eta > 2/a_{\max}$: past the wall	cut η (the classic $\div\, 10$)
fast drop, then endless crawl	$\kappa \gg 1$: ravine	**standardize features**; add **momentum**
slow large-period ringing	β too high: supertanker	lower β before touching η

Every cell is the quadratic story $x_{t+1} = (1 - \eta a)x_t$, read in a different corner of the (η, a, κ) space.

Lecture 17 — summary

- **GD derived** ⭐: trust L7′s line for one step — $\theta_{t+1} = \theta_t - \eta\nabla f$, payout $\approx \eta\|\nabla f\|^2$ (direction: L8′s Cauchy–Schwarz).
- **The dial**: crawl / converge / oscillate / diverge; the wall at $\eta = 2/a$ — two over the curvature.
- **Quadratics tell all** ⭐: iterates are $(1-\eta a)^t x_0$; per-axis factors on bowls; best $\eta = 1/a$.
- **The villain**: $\kappa = a_{\max}/a_{\min}$; best factor $(\kappa-1)/(\kappa+1) \implies$ bill $\propto \kappa$. Mystery 2 was $\kappa = 160$ in disguise.
- **Preconditioning**: curvature = feature scale squared — standardizing rounds the bowl.
- **Momentum**: $v = \beta v + g$; consistent directions $\times\, 1/(1-\beta)$, flip-flops damped — ravines tamed, not cured.

Lecture 17 — summary

Read before L18: MML §7.1 (gradient descent) · Distill, **Why Momentum Really Works** (assigned). **T8** this week: GD convergence on quadratics, condition numbers, and the optimizer race.

Practice problems

Try on paper; verify in the T8 notebook.

1. For $f(x) = 2x^2$ with $\eta = 0.4$: compute the contraction factor and the first three iterates from $x_0 = 1$. Converging?
2. On $f(x) = \frac{5}{2}x^2$: find every η that converges, the η that lands in one step, and what happens at exactly $\eta = 0.4$.
3. On $f(x, y) = \frac{1}{2}(x^2 + 16y^2)$ with the best single η: per-axis factors, and how many steps to shrink the slow-axis error $1000\,\times$?
4. Two features have standard deviations 1 and 50 (means 0). Estimate κ before and after standardization.
5. Momentum with $\beta = 0.95$: the effective memory length, and the limiting $|v|$ under alternating gradients $\pm g$.

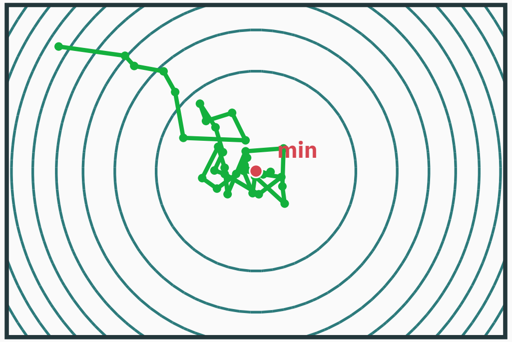
Practice problems

6. Reproduce Mystery 2 in NumPy ($a = 0.02, 3.2$, $\eta = 0.8$, 30 steps). Find the largest η that converges on **both**, and the steps problem A then needs for a $100\times$ error cut.

⭐⭐⭐ SGD exists

★ OPTIONAL

A real loss is an **average over data**: $\mathcal{L}(\theta) = \frac{1}{n}\sum_i \ell_i(\theta)$ — one true gradient costs n backprops. **Stochastic** GD steps on a random example's $\nabla \ell_i$ instead: right **on average**, noisy every step, $n\times$ cheaper.



That noise, minibatches, learning-rate schedules (see the `lr-schedule-visualizer` interactive), and the adaptive per-coordinate η of RMSProp/Adam — conditioning medicine, all of it — belong to **ES 335 and ES 667**. Today's contraction-factor story is the foundation they all stand on.

The learning rate is how far you trust a linear approximation.

Next: **Newton & Gauss–Newton** — what if you trusted a **quadratic** model instead?